KL DEEMED TO BE UNIVERSITY

PBL · PROGRAMMING IN JAVA · 2026

PROJECT BASED LEARNING — 26SC1101E PSPJ · SECTION 14 · TEAM 01

ATM *Simulator*

A console Java application that checks a PIN, handles deposits and withdrawals, and prints a mini statement.

pin → money → statement

01 ABSTRACT

The ATM Simulator authenticates the user with a PIN, then offers a menu to check balance, deposit, withdraw or view a mini statement. An Account class stores the name, PIN and balance, and a SavingsAccount subclass adds a minimum-balance rule. Invalid input and overdrawn withdrawals are handled with Java exceptions.

02 SYSTEM WORKFLOW

1. **PIN login** 3 attempts
2. **Menu** do-while loop
3. **Transaction** deposit or withdraw
4. **Statement** last 5 entries

`Account` ◁ `SavingsAccount`

`InsufficientBalanceException`

03 JAVA CONCEPTS USED

C01–C05 USED

- **C01** Data types, operators & Scanner input
- **C02** for, do-while, switch & if / else
- **C03** Methods, overloading & 1D array
- **C04** Classes, inheritance & overriding
- **C04** Encapsulation & custom exceptions
- **C05** String methods & String.format
- **C06** Collections & Streams: planned, not yet used

04 KEY FEATURES

only 3 tries!

- PIN login: 3 attempts, then the card is blocked
- Menu: balance, deposit, withdraw, statement, exit
- Rejects non-numeric input and amounts of 0 or less
- Savings account keeps a minimum balance of 500
- Custom exception for a failed withdrawal
- Mini statement of the last 5 transactions
- Masked account number on login (XXXXXX7890)

05 SAMPLE SESSION

row 4 gets rejected ↓

#	ACTION	INPUT	RESULT	BALANCE
1	Check balance	—	Current Balance	10000.0
2	Deposit	2000	Deposited successfully	12000.0
3	Withdraw	3000	Withdrawn successfully	9000.0
4	**Withdraw**	**8800**	**Rejected: minimum balance of 500**	**9000.0**

Starting balance 10000.0 on a savings account for Pushkal (XXXXXX7890).

06 SUBMITTED BY

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